Bitcoin Transaction Analysis: Legacy vs SegWit

Tool and Environment

Operating System: Windows
Bitcoin Core Version: 28.1 or higher
Compiler: GCC
Programming Language: C
JSON Parser: cJSON library
Mode Used: Regtest

Part 1 – Legacy P2PKH Transactions

Steps Performed

1. Created a wallet
2. Generated addresses
3. Generated blocks to obtain test coins
4. Created transactions
5. Decoded raw transactions

Addresses :

So the mapping is:

- **A address:** `mmVHKWaep44A45vSRBQLBaVEtt4Vtr9pxw`
- **B address:** `mzAhJmRR35aWzrxCMKXHm7aXPieUitDEam`
- **C address:** `msTceJSUH6CueFthjMtCC9rKQeTshey6Ck`
- **A → B txid:** `bcdc5f997dc19be1cb99fa11cf84c743a1c70a6621b2f2214f6522010b631ed8`
- **B → C txid:** `8279fba8b7367a878b00a5742c92f586aefa992e1f18f2f8cc0102109652922b`

Commands Used

- bitcoin-cli -regtest createwallet LabWallet
- bitcoin-cli -regtest -rpcwallet=LabWallet getnewaddress
- bitcoin-cli -regtest generatetoaddress 101 ADDRESS
- bitcoin-cli -regtest -rpcwallet=LabWallet sendtoaddress ADDRESS 10
- bitcoin-cli -regtest generatetoaddress 1 ADDRESS
- bitcoin-cli -regtest decoderawtransaction HEX

Transactions Created

- Transaction A → B
- Transaction B → C

Tx from A to B:


```
  "address": "mzAhJmRR35aWzrxCMKXHm7aXPieUitDEam",
  "parent_descs": [
    "pkh([4071e4f8/44h/1h/0h]tpubDCpkB9SoDXUHBQBJki3DRiQy1zbVVhTUoz4nMFoQUmpeDASFmdCsxJAhcb8tvKcqpf8TyhK9NkBBf3txcZ7UejFUXPvx7uERwDBp7iG2coA/0
*)#e2058m3d"
  ],
  "category": "receive",
  "amount": 9.99900000,
  "label": "",
  "vout": 0,
  "abandoned": false,
  "confirmations": 2,
  "blockhash": "3d6cedbd2d9a1f73206c157e091570d728e12c9a949704d5bd93327e7bdc41af",
  "blockheight": 104,
  "blockindex": 1,
  "blocktime": 1773142289,
  "txid": "bcdc5f997dc19be1cb99fa11cf84c743a1c70a6621b2f2214f6522010b631ed8",
  "wtxid": "bcdc5f997dc19be1cb99fa11cf84c743a1c70a6621b2f2214f6522010b631ed8",
  "walletconflicts": [
  ],
  "mempoolconflicts": [
  ],
  "time": 1773142271,
  "timereceived": 1773142271,
  "bip125-replaceable": "no"
},
```


Tx from B to C:


```
  "address": "msTceJSUH6CueFthjMtCC9rKQeTshey6Ck",
  "parent_descs": [
    "pkh([4071e4f8/44h/1h/0h]tpubDCpkB9SoDXUHBQBJki3DRiQy1zbVVhTUoz4nMFoQUmpeDASFmdCsxJAhcb8tvKcqpf8TyhK9NkBBf3txcZ7UejFUXPvx7uERwDBp7iG2coA/0
*)#e2058m3d"
  ],
  "category": "receive",
  "amount": 9.99800000,
  "label": "",
  "vout": 0,
  "abandoned": false,
  "confirmations": 1,
  "blockhash": "1f421c96f380703bd0f771d267188bb12dc6b3b35564587bdbc2b2909b29647a",
  "blockheight": 105,
  "blockindex": 1,
  "blocktime": 1773142712,
  "txid": "8279fba8b7367a878b00a5742c92f586aefa992e1f18f2f8cc0102109652922b",
  "wtxid": "8279fba8b7367a878b00a5742c92f586aefa992e1f18f2f8cc0102109652922b",
  "walletconflicts": [
  ],
  "mempoolconflicts": [
  ],
  "time": 1773142697,
  "timereceived": 1773142697,
  "bip125-replaceable": "no"
```

Legacy Transaction Structure:

- scriptSig (Legacy unlocking script):


```
"vin": [
  {
    "txid": "5e20e47b0a7866f5a430ac3990a57ffb5ef409c89f07ea2863517b46be464e75",
    "vout": 1,
    "scriptSig": {
      "asm": "30440220604abdae11116c89f4cade33694ebb91964b5c68a421259ab226855f90247d2102201808be973746fe18c3492b937f89c66fed48bd5cd4d32055b250600092266254[ALL] 0
      "hex": "4730440220604abdae11116c89f4cade33694ebb91964b5c68a421259ab226855f90247d2102201808be973746fe18c3492b937f89c66fed48bd5cd4d32055b250600092266254012 10
    },
    "sequence": 4294967293
  }
```


From output we can see the asm includes the signature + public key

scriptPubKey (locking script):


```
"vout": [
  {
    "value": 9.99900000,
    "n": 0,
    "scriptPubKey": {
      "asm": "OP_DUP OP_HASH160 cc958f3e40660b0e6d3d171d8e563a3983216379 OP_EQUALVERIFY OP_CHECKSIG",
      "desc": "addr(mzAhJmRR35aWzrxCMKXHm7aXPieUitDEam)#zkxvnwh8",
      "hex": "76a914cc958f3e40660b0e6d3d171d8e563a398321637988ac",
      "address": "mzAhJmRR35aWzrxCMKXHm7aXPieUitDEam",
      "type": "pubkeyhash"
    }
  }
```


OP_DUP OP_HASH160 <pubkeyhash> OP_EQUALVERIFY OP_CHECKSIG

The above line says, “To spend this bitcoin, you must provide a public key and a signature that matches this public key hash”.

Part 2 – SegWit P2SH-P2WPKH Transactions

Steps Performed:

1. Created SegWit wallet
2. Generated SegWit addresses
3. Generated blocks
4. Created SegWit transactions
5. Decoded the transactions

Commands used:

- bitcoin-cli -regtest createwallet SegWitWallet
- bitcoin-cli -regtest -rpcwallet=SegWitWallet getnewaddress "" p2sh-segwit
- bitcoin-cli -regtest generatetoaddress 101 ADDRESS
- bitcoin-cli -regtest -rpcwallet=SegWitWallet sendtoaddress ADDRESS 10
- bitcoin-cli -regtest generatetoaddress 1 ADDRESS
- bitcoin-cli -regtest decoderawtransaction HEX

Transactions created :

- Transaction A' → B'
- Transaction B' → C'

Addresses:

```
C:\Program Files\Bitcoin\daemon>bitcoin-cli -regtest -rpcwallet=SegWitWallet getnewaddress "" p2sh-segwit
2NGS3VmfquJqKuwXv3VTXqiZwC1ugbC6kVa

C:\Program Files\Bitcoin\daemon>bitcoin-cli -regtest -rpcwallet=SegWitWallet getnewaddress "" p2sh-segwit
2NEFGb65EHmK94M9QnBpnu6WADtFzg3kh1f

C:\Program Files\Bitcoin\daemon>bitcoin-cli -regtest -rpcwallet=SegWitWallet getnewaddress "" p2sh-segwit
2MtiVNisVpvm1EHQ11HpH7QtwuTszR2858d
```

Tx A' → B':


```
{
  "txid": "193bf05251609d4dd032412c80c274cff511e3ee19d4b07acfbf214775188445",
  "hash": "ddf74c8ec80d085b401fe20be7317164b93d8be525522bde2aba1266a9a98bf4",
  "version": 2,
  "size": 247,
  "vsize": 166,
  "weight": 661,
  "locktime": 206,
```

Tx B' → C':


```
{
  "txid": "5bff59b20e6d7117cff64e1ec61e48c912ebd732ed3dc68ddba9d70c3da621e2",
  "hash": "773e1797ec7ab5c906fb7ad52c19231918cb9238f4e1cd360326774990987f2b",
  "version": 2,
  "size": 247,
  "vsize": 166,
  "weight": 661,
  "locktime": 207,
  "vin": [
    {
      "txid": "193bf05251609d4dd032412c80c274cff511e3ee19d4b07acfbf214775188445",
      "vout": 1,
```


SegWit Transaction Structure:

- scriptSig → empty or minimal
- txinwitness → signature + public key
- scriptPubKey → P2SH script


```
  "vin": [
    {
      "txid": "193bf05251609d4dd032412c80c274cff511e3ee19d4b07acfbf214775188445",
      "vout": 1,
      "scriptSig": {
        "asm": "0014f31074eafb50cd720a73df3f4ef06f727e77424f",
        "hex": "160014f31074eafb50cd720a73df3f4ef06f727e77424f"
      },
      "txinwitness": [
        "3044022065d8f020c0563675f61158e8b7d1e1fe94943e985fed0aec5f96e3d86eec7ef1022037e1ae862b102db4a75a6e3eb6f38d8fc2fc9df7432a48290938cf9750fe884f01",
        "02932776d8ad11ce2b2511dffdca8d24553112af93471fbc9cd5fc51fb1eba9868"
      ],
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 9.99000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "OP_HASH160 a5ad64906d1aa73bd6b80fa1b0f327fa9f14de3d OP_EQUAL",
        "desc": "addr(2N8MFCUr5KfJmoTXY1aULGpYrGpwNswtBYG)#6dnurgm8",
        "hex": "a914a5ad64906d1aa73bd6b80fa1b0f327fa9f14de3d87",
        "address": "2N8MFCUr5KfJmoTXY1aULGpYrGpwNswtBYG",
        "type": "scripthash"
      }
    },
```

Part 3 – Analysis and Comparison

comparison table :

Transaction	Size	Vsize	Weight	Type
Legacy A→B	191	191	764	pubkeyhash
Legacy B→C	191	191	764	pubkeyhash
SegWit A′→B′	247	166	661	scripthash
SegWit B′→C′	247	166	661	scripthash

Explanation of Results

Why SegWit Transactions Are Smaller

SegWit moves signature data to the **witness section**, which is counted differently in block weight calculation.

Advantages:

- reduced virtual size
- lower transaction fees
- improved scalability
- fixes transaction malleability